

CHAPTER 1

Tang
&
Yang

Rong
Tang

Yun
Yang

2019–
2025

An-
nals
of
Sta-
tis-
tics
JRSS

MCMC

1.

1.1.

.

$$D$$

$$\frac{d \ll D}{D}$$

 D GAN
VAE

(1)

(2)

(3)

MCMC

Tang-
Yang

.

Tang
&
Yang
2019–
2025

MCMC

1

$\mathcal{Q}$
VAE

$\mathcal{Q}$

MCMC

2.

2.1.

.

$$\mathcal{M}^*$$

$$\mathbb{R}^D$$

$$d$$

$$\beta\text{-}$$

$$\beta \geq 1$$

$$P^*$$

$$\mathcal{M}^*$$

$$\mathrm{vol}_{\mathcal{M}^*}$$

$$p^*$$

$$\alpha-$$

$$\alpha \in [0,\beta-1]$$

$$p^*$$

$$\mathbb{R}^D$$

Lebesgue

————

Lebesgue

Par-
ti-
tion
of
Unity
Tang

$$(U_\lambda, \varphi_\lambda)$$

$$\{\gamma_\lambda\}$$

]]tang2022minimax

]]tang2023lowdim

Reach

reach

$$\text{reach}(\mathcal{M}) = \inf\{r > 0 : \mathcal{M}$$

r -

}

2.2.

IPM.

KL

Hellinger

Tang

IPM

$$d_{\mathcal{F}}(\mu, \nu) = \sup_{f \in \mathcal{F}} \left| \int_{\mathbb{R}^D} f(x) \, \mathrm{d}\mu - \int_{\mathbb{R}^D} f(x) \, \mathrm{d}\nu \right|.$$

$\mathcal{F}$ γ -
Hölder

Sobolev

1-
LipschitzWasserstein-
1

RKHS

MMD
 γ

trade-
off

$$\gamma = 1$$

$$d_\gamma$$

Wasserstein-
1

$$\gamma \rightarrow 0^+$$

$$d_\gamma$$

?]tang2025regression

2.3.

•
Tang

$$= n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}}.$$

$$n^{-1/2}$$

$$d$$

β -

d

Haus-
dorff

D

Lep-
ski

chain-
ing
peel-
ing
lo-
cal-
iza-
tion

Gir-
sanov

Poincaré

2.4.

**·
Gauss-
ian**

Langevin

$$p_{\sigma}(x)=\int p_{\text{data}}(y)\phi_{\sigma}(x-y)\,\mathrm{d}y$$

$$L_{\infty}$$

-

$$\nabla \log p_t(x)$$

$$t$$

—

$$t =$$

$$0$$

$$\frac{t}{T} =$$

?]tang2024diffusion

$$\mathcal{S}_{\text{NN}}$$

$$t$$

Girsanov

Gir-
sanov

SDE

3.

3.1. Tang
&
Yang
(2022)
—

.

Rong
Tang

and
Yun
Yang.
“Min-
i-
max
Rate
of
Dis-
tri-
bu-
tion
Es-
ti-
ma-
tion
on
Un-
known
Sub-
man-
i-
fold
un-
der
Ad-
ver-
sar-
ial
Losses.”
*An-
nals
of
Sta-
tis-
tics*,
2022.

 D

GAN
VAE

γ

$$n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}},$$

α

β

d

D

par-
ti-
tion
of
unity

VAE

Tay-
lor

Lep-
ski

Fano

Varshamov-
Gilbert

?]]tang2022minimax

?]]tang2023lowdim,
?]]tang2024diffusion

3.2. Tang
&
Yang
(2023a)

.

Rong
Tang
and
Yun
Yang.
“Es-
ti-
mat-
ing
Dis-
tri-
bu-
tions
with
Low-
dimensional
Struc-
tures
Us-
ing
Mix-
tures
of
Gen-
er-
a-
tive
Mod-
els.”
ICML,
2023.

VAE
GAN

MLD-
MAE

K

-

ρ_k

1-
Wasserstein

α -

β -

d

$$O\Big(n^{-\frac{\alpha\wedge(\beta-1)+1}{2(\alpha\wedge(\beta-1))+d}}\Big)\vee O\Big(\frac{\log n}{\sqrt{n}}\Big),$$

$$\begin{array}{l} \alpha\leq \\ \beta- \\ 1 \end{array}$$

/

?]tang2022minimax

?]tang2022minimax

3.3. ?]]tang2023robust

—

.

Rong

Tang,
Anir-
ban
Bhat-
tacharya,
Deb-
deep
Pati,
and
Yun
Yang.
“Ro-
bust
Bayesian
In-
fer-
ence
on
Rie-
mann-
ian
Sub-
man-
i-
fold,”
JRSS
B,
2023.

Stiefel

Un-
cer-
tainty
Quan-
tifi-
ca-
tion

Bayesian
RPE-
TEL

(1)

(2)

(3)

Bernstein-
von
Mises
(BvM)

set-
ting

Me-
trop-
o-
lis
(RRWM)

d

D

con-
duc-
tance
pro-
file

3.4. Tang
&
Yang
(2023c)
—

.

Rong
Tang

and
Yun
Yang.
“Min-
i-
max
Non-
para-
met-
ric
Two-
Sample
Test
un-
der
Ad-
ver-
sar-
ial
Losses,”
*AIS-
TATS*,
2023.

MMD

$O(n^2)$

Besov

Besov

1

ℓ_1
 ℓ_2

**Wasser-
stein**

1

2

$$O(n \log n + n^{2d/(4\alpha+d)})$$

MMD

$$O(n^2)$$

**3.5. Tang
&
Yang
(2024a)**

.

Rong
Tang
and
Yun
Yang.
“Adap-
tiv-
ity
of
Dif-
fu-
sion
Mod-
els
to
Man-
i-
fold
Struc-
tures,”
*AIS-
TATS*,
2024.

SOTA

Langevin

—

d D
Langevin**Gauss-
ian** $\tilde{O}(\sigma)$ σ n

d L_∞

—

1-
Wasserstein

$$\frac{1}{\sqrt{n}} \vee n^{-\frac{\alpha+1}{2\alpha+d}}.$$

??tang2022minimax

??tang2022minimax

3.6. [Metropolis-Adjusted Langevin](#)

Metropolis-
Adjusted
Langevin

.

Rong
Tang
and
Yun
Yang.
“Metropolis-
Adjusted
Langevin
Al-
go-
rithm:
Op-
ti-
mal

Di-
men-
sion
De-
pen-
dency
and
Be-
yond.”
ICML,
2024.

MALA

MCMC

$$O(d^{1/3})$$

s-
conductance
pro-
file

MALA

$O(d^{1/3})$

**burn-
in**

s-
conductance
pro-
file

con-
duc-
tance

Chen
et
al.
(2020)

con-
duc-
tance
pro-
file